

Table 1: Overall performance on ML-100K (best valid).

Model	NDCG@10	NDCG@20	Hit@10	Hit@20
IERec (baseline)	0.0540	0.0746	0.1141	0.1965
IERec + EGPF (ours)	0.0549	0.0760	0.1161	0.2003
Δ (abs.)	+0.0009	+0.0014	+0.0020	+0.0038
Δ (%)	+1.67%	+1.88%	+1.75%	+1.93%

Table 2: Validation performance on entropy-based subsets (fixed split by baseline quantiles: low/medium/high = 30%/40%/30%).

Subset	Model	NDCG@10	NDCG@20	Hit@10	Hit@20
Low	IERec (baseline)	0.0852	0.1169	0.1813	0.3082
	IERec + EGPF (ours)	0.0864	0.1191	0.1839	0.3138
	Δ (abs.)	+0.0012	+0.0022	+0.0026	+0.0056
	Δ (%)	+1.46%	+1.85%	+1.43%	+1.82%
Medium	IERec (baseline)	0.0517	0.0709	0.1088	0.1853
	IERec + EGPF (ours)	0.0511	0.0726	0.1078	0.1935
	Δ (abs.)	-0.0006	+0.0017	-0.0010	+0.0083
	Δ (%)	-1.25%	+2.42%	-0.95%	+4.46%
High	IERec (baseline)	0.0295	0.0425	0.0622	0.1140
	IERec + EGPF (ours)	0.0315	0.0432	0.0645	0.1106
	Δ (abs.)	+0.0020	+0.0007	+0.0023	-0.0035
	Δ (%)	+6.82%	+1.74%	+3.70%	-3.03%